

# Drifts and currents in SOLPS

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## Outline

1. New terms in the equations
2. Modified solution procedure
3. Associated code variables and switches
4. Running with drifts

## Equations : New terms

Since we are solving equations of the form  $\nabla(\dots) = S$ , we only need consider the *divergent* contributions to the drift flows, while other terms are treated as sources.  $\perp$  is the coordinate in the diamagnetic direction. The terms added are referred to with indices as follows:

(a) anomalous (includes  $\mathbf{E} \times \mathbf{B}$  contribution)

(in) inertial and gyroviscous force

(vis) viscous force

(s) ion-neutral friction

(dia) *divergent* diamagnetic contribution

The (vis) and (in) terms are piecewise replacements of parts of the 21-moment model already existing in **SOLPS**. The 21-moment model is turned on with the switches `b2trcl_lthf21` for the thermal force treatment and `b2trcl_lvis21` for the viscous force. The code checks that not both viscosity models are used simultaneously.

# Continuity equation

$$\frac{\partial n}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} n \left( b_x V_{\parallel} + b_z V_{\perp}^{(0)} \right) \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} n V_y^{(0)} \right) = S^n \quad (1)$$

$$V_{\perp} = V_{\perp}^{(a)} - \frac{1}{enB} \frac{1}{h_y} \frac{\partial n T_i}{\partial y} + V_{\perp}^{(in)} + \tilde{V}_{\perp}^{(vis)} + V_{\perp}^{(s)} \quad (2)$$

$$V_y = V_y^{(a)} + \frac{b_z}{enB} \frac{1}{h_x} \frac{\partial n T_i}{\partial x} + V_y^{(in)} + \tilde{V}_y^{(vis)} + V_y^{(s)} \quad (3)$$

$$V_{\perp}^{(0)} = V_{\perp}^{(a)} + V_{\perp}^{(in)} + \tilde{V}_{\perp}^{(vis)} + V_{\perp}^{(s)} + \tilde{V}_{\perp}^{(dia)} \quad (4)$$

$$V_y^{(0)} = V_y^{(a)} + V_y^{(in)} + \tilde{V}_y^{(vis)} + V_y^{(s)} + \tilde{V}_y^{(dia)} \quad (5)$$

$$V_{\perp}^{(a)} = -\frac{1}{B} \frac{1}{h_y} \frac{\partial \Phi}{\partial y} - \frac{D}{T_e + T_i} \frac{b_z}{h_x} \left( \frac{\partial p}{n \partial x} - \frac{3}{2} \frac{\partial T_e}{\partial x} \right) \quad (6)$$

$$-D_{AN}^n \frac{1}{h_x n} \frac{\partial n}{\partial x} - D_{AN}^p \frac{1}{h_x n} \frac{\partial p_i}{\partial x} \quad (7)$$

$$V_y^{(a)} = -\frac{b_z}{B} \frac{1}{h_x} \frac{\partial \Phi}{\partial x} - \frac{D}{T_e + T_i} \frac{1}{h_y} \left( \frac{\partial p}{n \partial y} - \frac{3}{2} \frac{\partial T_e}{\partial y} \right) \quad (8)$$

$$-D_{AN}^n \frac{1}{h_y n} \frac{\partial n}{\partial y} - D_{AN}^p \frac{1}{h_y n} \frac{\partial p_i}{\partial y} \quad (9)$$

$$\vec{V}^{(in)} = \vec{j}^{(in)} / en, \quad \vec{V}^{(vis)} = \vec{j}^{(vis)} / en, \quad \vec{V}^{(s)} = \vec{j}^{(s)} / en \quad (10)$$

$$\tilde{V}_{\perp}^{(dia)} = \frac{T_i B_z}{e b_z} \frac{\partial}{h_y \partial y} \left( \frac{1}{B^2} \right), \quad \tilde{V}_y^{(dia)} = -\frac{T_i B_z}{e} \frac{\partial}{h_x \partial x} \left( \frac{1}{B^2} \right) \quad (11)$$

# Potential equation

$$\frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} \tilde{j}_x \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} \tilde{j}_y \right) = 0 . \quad (12)$$

$$\vec{j} = \vec{j}_{\parallel} + \vec{j}^{(AN)} + \vec{j}^{(dia)} + \vec{j}^{(vis)} + \vec{j}^{(in)} + \vec{j}^{(s)} . \quad (13)$$

$$\tilde{j}_x = b_z \tilde{j}_{\perp} + b_x \tilde{j}_{\parallel} , \quad \vec{j}^{(AN)} = -\sigma^{(AN)} \nabla \Phi = -k^{(AN)} e n \nabla \Phi . \quad (14)$$

$$j_{\parallel} = \sigma_{\parallel} \left[ \frac{b_x}{e} \frac{1}{h_x} \left( \frac{\partial n T_e}{n \partial x} + 0.71 \frac{\partial T_e}{\partial x} \right) - \frac{b_x}{h_x} \frac{\partial \Phi}{\partial x} \right] . \quad (15)$$

$$\tilde{j}_{\perp}^{(dia)} = \frac{1}{b_z} \frac{n (T_e + T_i) B_z}{h_y} \frac{\partial}{\partial y} \left( \frac{1}{B^2} \right) , \quad \tilde{j}_y^{(dia)} = - \frac{n (T_e + T_i) B_z}{h_x} \frac{\partial}{\partial x} \left( \frac{1}{B^2} \right) \quad (16)$$

$$\vec{j}^{(vis)} = \vec{j}^{(vis\parallel)} + \vec{j}^{(vis\perp)} + \vec{j}^{(visq)} \quad (17)$$

$$\tilde{j}_{\perp}^{(vis\parallel)} = - \frac{B_x \eta_0}{3 \sqrt{B}} \frac{\partial (V_{\parallel} \sqrt{B})}{h_x \partial x} \frac{\partial}{h_y \partial y} \left( \frac{1}{B^2} \right) \quad (18)$$

$$\tilde{j}_y^{(vis\parallel)} = b_z \frac{B_x \eta_0}{3 \sqrt{B}} \frac{\partial (V_{\parallel} \sqrt{B})}{h_x \partial x} \frac{\partial}{h_x \partial x} \left( \frac{1}{B^2} \right) \quad (19)$$

## More small but complicated terms

The current produced by components of viscosity tensor connected with heat fluxes is:

$$\tilde{j}_{\perp}^{(visq)} = -\frac{0.24B_x}{\sqrt{B}\nu_{ii}} \frac{\partial \left( q_{i\parallel}^{(0)} \sqrt{B} \right)}{h_x \partial x} \frac{\partial}{h_y \partial y} \left( \frac{1}{B^2} \right) \quad (20)$$

$$\tilde{j}_y^{(visq)} = b_z \frac{0.24B_x}{\sqrt{B}\nu_{ii}} \frac{\partial \left( q_{i\parallel}^{(0)} \sqrt{B} \right)}{h_x \partial x} \frac{\partial}{h_x \partial x} \left( \frac{1}{B^2} \right) \quad (21)$$

The contribution from perpendicular viscosity is

$$\begin{aligned} j_{\perp}^{(vis\perp)} = & \frac{1}{B} \left[ \frac{1}{h_x} \frac{\partial}{\partial x} \left( \eta_1 \frac{\partial V_y}{h_x \partial x} \right) + \frac{1}{h_y} \frac{\partial}{\partial y} \left( \eta_1 \frac{\partial V_y}{h_y \partial y} \right) \right. \\ & \left. + \frac{1}{h_x h_y} \frac{\partial \eta_1}{\partial x} \frac{\partial V_{\perp}}{\partial y} - \frac{1}{h_x h_y} \frac{\partial \eta_1}{\partial y} \frac{\partial V_{\perp}}{\partial x} \right] + \frac{4b_x}{B} \frac{1}{h_x h_y} \left( \frac{\partial \eta_1}{\partial x} \frac{\partial V_{\parallel}}{\partial y} - \frac{\partial \eta_1}{\partial y} \frac{\partial V_{\parallel}}{\partial x} \right) \end{aligned} \quad (22)$$

$$\begin{aligned} j_y^{(vis\perp)} = & -\frac{1}{B} \left[ \frac{1}{h_y} \frac{\partial}{\partial y} \left( \eta_1 \frac{\partial V_{\perp}}{h_y \partial y} \right) + \frac{1}{h_x} \frac{\partial}{\partial x} \left( \eta_1 \frac{\partial V_{\perp}}{h_x \partial x} \right) \right. \\ & \left. - \frac{1}{h_x h_y} \frac{\partial \eta_1}{\partial x} \frac{\partial V_y}{\partial y} + \frac{1}{h_x h_y} \frac{\partial \eta_1}{\partial y} \frac{\partial V_y}{\partial x} \right]. \end{aligned} \quad (23)$$

where  $\eta_3 = \frac{nT_i}{2\omega_{ci}}$ ,  $\eta_2 = nm_i D_{AN}^i$ ,  $\eta_1 = \frac{\eta_2}{4}$ .

# Inertial force and ion-neutral friction

The current caused by inertial forces is:

$$\begin{aligned} j_{\perp}^{(in)} = & -\frac{m_i}{B} \left[ \frac{\partial n V_y}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} n \left( V_{\perp}^{(0)} b_z + V_{\parallel} b_x \right) V_y \right) \right. \\ & \left. + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} n V_y^{(0)} V_y \right) \right] - \frac{2b_x}{B} \frac{1}{h_x} \frac{\partial}{\partial x} \left( \eta_3 \frac{\partial V_{\parallel}}{h_x \partial x} \right) + \frac{m_i}{B} n V_{\parallel}^2 \frac{\partial \ln h_z}{h_y \partial y} \end{aligned} \quad (24)$$

$$\begin{aligned} j_y^{(in)} = & \frac{m_i}{B} \left[ \frac{\partial n V_{\perp}}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} n \left( V_{\perp}^{(0)} b_z + V_{\parallel} b_x \right) V_{\perp} \right) \right. \\ & \left. + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} n V_y^{(0)} V_{\perp} \right) \right] - \frac{2b_x}{B} \frac{1}{h_x} \frac{\partial}{\partial x} \left( \eta_3 \frac{\partial V_{\parallel}}{h_y \partial y} \right) - \frac{m_i}{B} n V_{\parallel}^2 \frac{\partial \ln h_z}{h_x \partial x} \end{aligned} \quad (25)$$

$$j_{\perp}^{(s)} = \frac{S_{iy}^m}{B}, \quad j_y^{(s)} = -\frac{S_{i\perp}^m}{B}, \quad (26)$$

where  $S_{iy}^m$ ,  $S_{i\perp}^m$  are the momentum sinks due to ion-neutral friction.

$$S_{i\perp}^m = -n \mu_{iN} \nu_{iN} (V_{\perp} - V_{\perp N}), \quad S_{iy}^m = -n \mu_{iN} \nu_{iN} (V_y - V_{yN}), \quad (27)$$

where  $\mu_{iN} = m_i/2$  is the effective mass,  $\nu_{iN} = n_N \langle V_{iN} \sigma_{cx} \rangle$  is the ion-neutral collision frequency,  $n_N$  is the neutral density and  $\vec{V}_N$  is the neutral mean velocity. For the ion velocity used above, only electric and diamagnetic drifts are left, neglecting smaller terms, to obtain

$$\tilde{j}_x^{(s)} = \tilde{j}_{\perp}^{(s)} b_z = -\sigma_{iN} b_z^2 \frac{\partial \Phi}{h_x \partial x} - \sigma_{iN} b_z^2 \frac{1}{en} \frac{\partial n T_i}{h_x \partial x} + \sigma_{iN} B_z V_{yN}, \quad (28)$$

$$\tilde{j}_y^{(s)} = -\sigma_{iN} \frac{\partial \Phi}{h_y \partial y} - \sigma_{iN} \frac{1}{en} \frac{\partial n T_i}{h_y \partial y} - \sigma_{iN} B V_{\perp N}, \quad (29)$$

where  $\sigma_{iN}$  is the ion-neutral perpendicular conductivity

$$\sigma_{iN} = \frac{n m_i \langle V_{iN} \sigma_{cx} \rangle n_N}{2B^2}. \quad (30)$$

# Electron energy balance equation

$$\begin{aligned}
 & \frac{3}{2} \frac{\partial n T_e}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \right) \\
 & + \frac{n T_e}{\sqrt{g}} \frac{\partial}{\partial x} \left\{ \frac{\sqrt{g}}{h_x} \left[ (V_{\parallel} - j_{\parallel}/en) b_x - \frac{D}{T_e + T_i} \frac{b_z^2}{h_x} \left( \frac{\partial p}{n \partial x} - \frac{3}{2} \frac{\partial T_e}{\partial x} \right) \right. \right. \\
 & \quad \left. \left. - D_{AN}^n \frac{1}{h_x n} \frac{\partial n}{\partial x} - D_{AN}^p \frac{1}{h_x n} \frac{\partial p_i}{\partial x} \right] \right\} \\
 & + n T_e \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left\{ \frac{\sqrt{g}}{h_y} \left[ - \frac{D}{T_e + T_i} \frac{1}{h_y} \left( \frac{\partial p}{n \partial y} - \frac{3}{2} \frac{\partial T_e}{\partial y} \right) \right. \right. \\
 & \quad \left. \left. - D_{AN}^n \frac{1}{h_y n} \frac{\partial n}{\partial y} - D_{AN}^p \frac{1}{h_y n} \frac{\partial p_i}{\partial y} \right] \right\} \\
 & = Q_e + n T_e B \frac{1}{h_x h_y} \left[ \frac{\partial \Phi}{\partial y} \frac{\partial}{\partial x} \left( \frac{1}{B^2} \right) - \frac{\partial \Phi}{\partial x} \frac{\partial}{\partial y} \left( \frac{1}{B^2} \right) \right] \quad (31)
 \end{aligned}$$

where

$$Q_e = -Q_{\Delta} - Q_u, \quad Q_{\Delta} = \frac{3m_e}{m_i} n \nu_{ei} (T_e - T_i), \quad Q_u = R_{ie\parallel} j_{\parallel}/en \quad (32)$$

# Electron heat fluxes

The heat fluxes are:

$$\begin{aligned} \tilde{q}_{ex} = & \frac{3}{2} n T_e \left( b_z V_{\perp}^{(a)} + b_x V_{\parallel} - b_x j_{\parallel} / e n \right) - 0.71 b_x j_{\parallel} T_e / e - \kappa_{e\parallel} \frac{b_x^2}{h_x} \frac{\partial T_e}{\partial x} \\ & - \kappa_{e\perp} b_z^2 \frac{1}{h_x} \frac{\partial T_e}{\partial x} + \frac{3}{2} \frac{T_e}{e B} \frac{\nu_{ei}}{\omega_{ce}} \frac{b_z^2}{h_x} \frac{\partial n (T_e + T_i)}{\partial x} - \frac{5}{2} n T_e^2 \frac{B_z}{e} \frac{\partial}{h_y \partial y} \left( \frac{1}{B^2} \right) \end{aligned} \quad (33)$$

$$\begin{aligned} \tilde{q}_{ey} = & \frac{3}{2} n T_e V_y^{(a)} - \kappa_{e\perp} \frac{1}{h_y} \frac{\partial T_e}{\partial y} + \frac{3}{2} \frac{T_e}{e B} \frac{\nu_{ei}}{\omega_{ce}} \frac{1}{h_y} \frac{\partial n (T_e + T_i)}{\partial y} \\ & + \frac{5}{2} n T_e^2 \frac{B_z}{e} \frac{\partial}{h_x \partial x} \left( \frac{1}{B^2} \right) \end{aligned} \quad (34)$$

# Ion energy balance equation

$$\begin{aligned}
 & \frac{3}{2} \frac{\partial n T_i}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} \tilde{q}_{ix} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \right) \\
 & + \frac{n T_i}{\sqrt{g}} \frac{\partial}{\partial x} \left\{ \frac{\sqrt{g}}{h_x} \left[ V_{\parallel} b_x - \frac{D}{T_e + T_i} \frac{b_z^2}{h_x} \left( \frac{\partial p}{n \partial x} - \frac{3}{2} \frac{\partial T_e}{\partial x} \right) \right. \right. \\
 & \quad \left. \left. - D_{AN}^n \frac{1}{h_x n} \frac{\partial n}{\partial x} - D_{AN}^p \frac{1}{h_x n} \frac{\partial p_i}{\partial x} \right] \right\} \\
 & + n T_i \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left\{ \frac{\sqrt{g}}{h_y} \left[ - \frac{D}{T_e + T_i} \frac{1}{h_y} \left( \frac{\partial p}{n \partial y} - \frac{3}{2} \frac{\partial T_e}{\partial y} \right) \right. \right. \\
 & \quad \left. \left. - D_{AN}^n \frac{1}{h_y n} \frac{\partial n}{\partial y} - D_{AN}^p \frac{1}{h_y n} \frac{\partial p_i}{\partial y} \right] \right\} \\
 & = Q_{\Delta} - \frac{\eta_0}{3} \left( 2 b_x \frac{\partial V_{\parallel}}{h_x \partial x} \right)^2 \\
 & \quad + n T_i B \frac{1}{h_x h_y} \left[ \frac{\partial \Phi}{\partial y} \frac{\partial}{\partial x} \left( \frac{1}{B^2} \right) - \frac{\partial \Phi}{\partial x} \frac{\partial}{\partial y} \left( \frac{1}{B^2} \right) \right] \quad (35)
 \end{aligned}$$

where

$$\tilde{q}_{ix} = \frac{3}{2} n T_i \left( b_z V_{\perp}^{(a)} + b_x V_{\parallel} \right) - \kappa_{i\parallel} \frac{b_x^2}{h_x} \frac{\partial T_i}{\partial x} - \kappa_{i\perp} b_z^2 \frac{1}{h_x} \frac{\partial T_i}{\partial x} + \frac{5}{2} n T_i b_z \tilde{V}_{\perp}^{(dia)} \quad (36)$$

$$\tilde{q}_{iy} = \frac{3}{2} n T_i V_y^{(a)} - \kappa_{e\perp} \frac{1}{h_y} \frac{\partial T_i}{\partial y} + \frac{5}{2} n T_i \tilde{V}_y^{(dia)} \quad (37)$$

# Parallel momentum equation

$$\begin{aligned}
 m_i & \left[ \frac{\partial n V_{\parallel}}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} n \left( V_{\perp}^{(0)} b_z + V_{\parallel} b_x \right) V_{\parallel} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} n V_y^{(0)} V_{\parallel} \right) \right] \\
 &= - \frac{b_x}{h_x} \frac{\partial n T_i}{\partial x} - b_x \frac{en}{h_x} \frac{\partial \Phi}{\partial x} + F_k + \frac{4}{3} b_x B^{3/2} \frac{\partial}{h_x \partial x} \left( \frac{\eta_0 b_x}{B^2} \frac{\partial (\sqrt{B} V_{\parallel})}{h_x \partial x} \right) \\
 & \quad + B^{3/2} b_x \frac{\partial}{h_x \partial x} \left( \frac{b_x}{\nu_{ii} B^2} \frac{\partial (\sqrt{B} q_{i\parallel}^{(0)})}{h_x \partial x} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y^2} \eta_2 \frac{\partial V_{\parallel}}{\partial y} \right) \\
 & \quad + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x^2} \eta_2 \frac{\partial V_{\parallel}}{\partial x} \right) S_{i\parallel}^m + R_{ie\parallel} \quad (38)
 \end{aligned}$$

where the Coriolis force term is:

$$F_k = -m_i \left[ \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left( \frac{\sqrt{g}}{h_x} n V_{\parallel} \tilde{V}_{\perp}^{(dia)} b_z \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left( \frac{\sqrt{g}}{h_y} n V_{\parallel} \tilde{V}_y^{(dia)} \right) \right] \quad (39)$$

## New solution procedure

In the standard setup, we solve the B2 equations in the following order:

$\Rightarrow$  momentum  $\Rightarrow$  continuity  $\Rightarrow$  potential  $\Rightarrow$  energy

With drifts, the sheath boundary condition becomes:

$$V_{\parallel} \geq C_s - \frac{b_z}{b_x} V_{\perp} > 0 \quad (40)$$

$$j_x = en \left( b_x C_s - b_x \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\Phi}{T_e}\right) (1 - \gamma) \right) \quad (41)$$

For very small pitch angles ( $b_x \ll b_z$ ), the  $V_{\perp}$  term can easily overcome  $C_s$  and lead to negative  $V_{\parallel}$ . To avoid such an unphysical plasma flow exiting the plate (denoted **wrong\_flow**), the boundary condition routine **b2stbc** now tests that  $V_{\parallel} > 0$ . The call to the sources routine **b2srsl** within the solution loop is controlled by the switch **b2news\_recalculate\_contributions**.  $\gamma$  is the secondary electron emission coefficient (default value 0.5, results from a kinetic model can be used with the **b2stbc\_secmodel** switch). The solution procedure is modified to be:

```
wrong_flow = .false.; ipass = 0
START solution loop
  DO IS = 0, NS-1
    solve momentum equation for species IS
    solve continuity equation for species IS
  END DO
  CALL B2SRAL (which calls B2STBC and computes wrong_flow)
  IF (.not.wrong_flow) solve for potential equation
  solve for energy equations
  ipass = ipass + 1
  IF (ipass > potit) give up!
  IF (wrong_flow) go back to START
END solution loop
```

## New code variables

All the new variables below are to be found in the module **b2mod\_plasma**. The index 0 refers to the projection onto the *poloidal* direction, the index 1 onto the *radial* direction.

**vadia**(-1:nx,-1:ny,0:1,0:ns-1) diamagnetic velocity  $\tilde{V}^{(dia)}$

**vaecrb**(-1:nx,-1:ny,0:1,0:ns-1)  $\mathbf{E} \times \mathbf{B}$  velocity

**uadia**(-1:nx,-1:ny,0:1,0:ns-1) sum of the above

**fchin**(-1:nx,-1:ny,0:1) inertial current  $j^{(in)}$

**fchvispar**(-1:nx,-1:ny,0:1) parallel viscosity current  $\tilde{j}^{(vis\perp)}$

**fchvisq**(-1:nx,-1:ny,0:1) heat flux viscosity current  $\tilde{j}^{(visq)}$

**fch\_p**(-1:nx,-1:ny) parallel current  $j_{\parallel}$

**fchdia**(-1:nx,-1:ny,0:1) diamagnetic current  $\tilde{j}^{(dia)}$

**fna\_nodrift**(-1:nx,-1:ny,0:1,0:ns-1) Particle flow without drifts included

**csigin**(-1:nx,-1:ny,0:1) Ion-neutral friction electrical conductivity

**csig\_an**(-1:nx,-1:ny,0:1) Anomalous electrical conductivity

The following are available in **b2plot**:

**ud** uadia(:, :, 0, :)

**vd** uadia(:, :, 1, :)

**cd** fchdia(:, :, 0)

**dd** fchdia(:, :, 1)

## New switches

### ALL SWITCHES ARE DOCUMENTED IN B2CDCI.F

There are two families of master switches:

|                        |   |                      |
|------------------------|---|----------------------|
| b2news_facdrift_start  | { | b2news_facExB_start  |
| b2news_facdrift_target |   | b2news_facExB_target |
| b2news_facdrift_inc    |   | b2news_facExB_inc    |
| b2news_facdrift_dec    |   | b2news_facExB_dec    |
| OR b2news_ExB          |   |                      |

**\_start** initial value of parameter

**\_target** desired value of parameter

**\_inc** multiplicative increment of parameter on the timestep

**\_dec** multiplicative decrement of parameter if the timestep fails to converge

Related switches (those in **green** are subservient to **facdrift**):

**b2sdia\_facgt** Multiplies the  $\nabla T \times \mathbf{B}$  source term in energy equations

**b2stbc\_fchy\_dia** Multiplies the neoclassical current boundary condition:

$$\left( \text{surface-averaged current} \sim \frac{\partial p}{\partial r} \right)$$

**b2stbc\_neoclassical** Turns on the neoclassical current boundary condition when  $\text{facdrift} = 0$

**b2stbc\_fchy\_dia\_coreonly** Only uses the neoclassical current boundary condition on the core boundary

**b2tfhe\_vispar** Multiplies parallel viscosity terms

**b2tfhe\_visq** Multiplies heat flux viscosity terms

**b2tfhe\_neutral** Multiplies ion-neutral friction terms

**b2news\_iy\_nocoreExB** Turns off  $\mathbf{E} \times \mathbf{B}$  flows for core cells where  $iy \leq iy_{nocoreExB}$

## More new switches

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Other new switches of interest:

**b2stbc\_integral\_current** Attempts to set current to zero on SOL and PFR boundaries

**b2tfhe\_anomalous** Turns on radial anomalous conductivity term

**b2news\_poteq** Turns on potential equation

**b2tfhe\_no\_current** Turns off all currents

**b2news\_recalculate\_contributions** Turns on check of `wrong_flow` during solution loop

**b2stbc\_secmodel** Computes the secondary electron emission coefficient  $\gamma$  from results of a kinetic model

## Setting up a case with drifts

1. Obtain a converged run without drifts
2. Set **facdrift\_start** (and/or **facExB\_start**) and **\_target** to the same small value (e.g.: 0.01). Alternatively, directly set **b2news\_ExB**.
3. Let run for a few hundred iterations
4. Increase **\_target** values
5. Continue the run
6. Restart run as necessary setting respective new **\_start** values to the current **facdrift** and **facExB** values
7. Lower time-step if necessary
8. Lower numerical perpendicular conductivity parameters **cf<sub>sig</sub>** and **cf<sub>alf</sub>** *simultaneously*, as much as possible while still avoiding numerical instabilities. This reduces the values of  $k^{(AN)}$  so as to get back the purely classical character of the current continuity equation, where the radial current is now carried not by an anomalous term, but only by the classical terms (including drifts).

# Reversing field and/or plasma current

It is often useful to revert magnetic field (changing  $b_z$  from +1 to -1) and/or plasma current (changing the sign of  $b_x$ ) when studying the effects of drifts. This can now be done using the switches below:

1. In a new run directory, copy over **b2ag.dat**

2. Set as needed:

**b2agfs\_Bt\_reversal** 0: usual  $B_T$  direction, 1: reversed  $B_T$

**b2agfs\_pit\_rescale** multiplies the poloidal field (i.e.  $I_p$ )

**b2agfs\_Bt\_rescale** multiplies the entire field array

3. Both multipliers can be set to **-1.0** for reversal

4. Obtain new **b2fgmtry** file with **b2run b2ag** command

5. With the new **b2fgmtry**, restart the run!

A way to get back (almost) to SOLPS4.0

Set

```
b2news_poteq = 0
```

```
b2tfhe_no_current = 1
```

```
b2mndr_BoRiS = 1.0
```

This turns off the potential equation, zeroes out all currents, and goes back to solving  $T_e$  and  $T_i$  equations instead of total energy equations.